

DOCUMENT 01 · REQUIREMENTS BASELINE v2.0

SALESTORM

Requirements & Assumptions

Industry-grade specification for a high-scale e-commerce flash-sale platform

10,000

CONCURRENT BUY
ATTEMPTS

100

UNITS OF
STOCK

500K

PEAK EDGE
REQUESTS / SEC

0

OVERSELLS /
DUPLICATE CHARGES

Document control

System	SALESTORM E-Commerce Flash-Sale Platform
Version / Status	v2.0 — Industry-grade baseline (supersedes v1.0). Architecture decisions remain deferred to 02_HLD.
Source basis	SALESTORM SYSCRAFTERS 2026 Hackathon Brief (19 pages) plus industry practice for flash-sale, payment and marketplace systems.
Numeric values	Where the brief gives no number, values are labelled Team Assumption (TA) and are tunable. Brief-given values are labelled Brief .
Next artifact	02_HLD — answers the 14 handoff questions in Section 22.

WHAT CHANGED IN V2.0

v1.0 faithfully restated the brief. v2.0 turns it into a testable specification: quantified NFRs, a derived capacity model, actors and acceptance criteria, reservation-timing rules, fairness and anti-bot controls, payment-industry requirements (PCI, 3-D Secure, webhooks, reconciliation), split reservation/order state machines, DR targets, ops requirements, compliance, a risk register and a requirement-to-verification matrix.

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How to Read This Document

Conventions, ID scheme, priorities and verification codes

Prefix	Domain	Prefix	Domain
FR	Functional requirements	MSG	Messaging and event reliability
NFR	Non-functional (quantified)	SEC	Security and compliance
RSV / INV	Reservation and inventory	OBS / OPS	Observability and operations
TRF	Traffic protection and fairness	DR	Availability and disaster recovery
PAY	Payments	DAT	Data governance
ORD	Orders and fulfilment	US / G / R	User story / Guarantee / Risk

Priority: MUST = release-blocking and defended at jury; SHOULD = expected in an industry design; COULD = stretch.
Verify: T = automated test · S = simulation/harness · L = load test · F = failure injection · R = design review · M = production monitoring · D = demo walkthrough.

Backward compatibility: all v1.0 IDs (FR-01...18, NFR-01...14, A-01...08, C-01...09) are retained. NFR-02, NFR-08 and NFR-09 were duplicates of guarantees; they are kept for traceability and now cross-reference the canonical item (G-xx / RSV-xx / PAY-xx).

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Problem Statement & Business Objectives

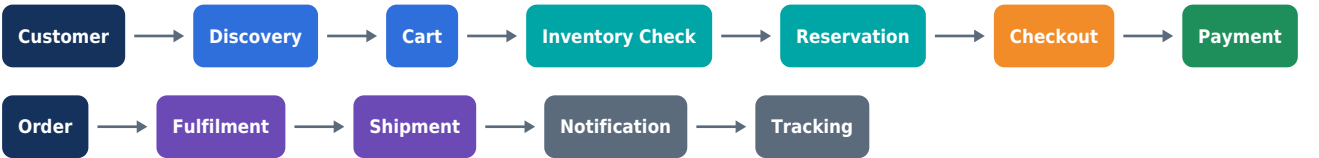
Why SALESTORM exists and what success looks like in numbers

SALESTORM models a limited-stock flash sale. A sudden surge of customers selects **Buy Now** on the same product within seconds, while payment gateways, databases and downstream services may slow down or fail. The critical scenario is **Product X: 100 units, 10,000 simultaneous customers**. The platform must decide, deterministically and fairly, who wins stock; never sell more than exists; and keep payment and order state correct through failures.

CENTRAL BUSINESS QUESTION

How can thousands of simultaneous purchase requests for limited inventory be handled without overselling, while keeping payment and order processing reliable?

1.1 Required business flow



1.2 Business objectives with measurable outcomes

ID	Objective	Measurable outcome (Brief / TA)	Priority
BO-01	Survive sudden traffic without collapse	Peak 500,000 req/s at edge (Brief); error rate < 0.1% 5xx (TA)	MUST
BO-02	Never oversell	Successful sales ≤ 100 for 100 units; conservation invariant holds 100% of the time	MUST
BO-03	Temporary reservation with expiry	Unpaid reservations released within TTL + 30 s (TA)	MUST
BO-04	Safe, idempotent payment	0 duplicate charges across 100% of retries and webhooks	MUST
BO-05	Consistent, traceable order lifecycle	100% of captured payments reach a terminal order state; full audit trail	MUST
BO-06	Graceful recovery from failures	RPO = 0, RTO ≤ 5 min single-AZ (TA); paid-not-ordered converted ≤ 60 s after recovery	MUST
BO-07	Fair, bot-resistant sale	Per-customer limit enforced; bot traffic blocked at edge; admission control active	SHOULD
BO-08	Buildable and operable by a team	Runbooks, SLOs, dashboards and kill switch defined before launch	SHOULD

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Scope, Actors & Stakeholders

Who uses the system, what is in, what is deliberately out

2.1 Actors

Actor	Type	Goals and interactions
Customer	Primary human	Browse, add to cart, Buy Now, pay, track order, cancel, receive notifications
Merchandiser / Sale Admin	Internal human	Configure sale, stock, price, per-customer limits, TTL; pause sale; force sold-out (RBAC, audited)
Payment Gateway	External system	Authorise/capture/refund; sends async webhooks; provides settlement reports
Logistics / Delivery Partner	External system	Receives shipment requests; returns tracking events
Notification Providers	External system	Email / SMS / push delivery
Identity Provider	External system	Authenticates customers; issues tokens
Ops / SRE	Internal human	Monitors SLOs, handles alerts, runs runbooks, replays DLQ, executes DR
Finance / Reconciliation	Internal human	Reviews daily settlement matching and unmatched items
Fraud / Bot Actor	Adversary	Automated purchasing, credential stuffing, replay, scalping (threat actor)

2.2 In scope vs. out of scope

IN SCOPE

- Customer identity, catalogue, categories, sale/deal/coupon rules
- Cart, inventory, reservation, checkout
- Payment processing, refund, reconciliation
- Order lifecycle, cancellation, recovery
- Fulfilment, shipment, delivery tracking, notifications
- Reliability, security, scalability, observability of the purchase path

OUT OF SCOPE (declared)

- Recommendation / search relevance engine
- Seller onboarding and marketplace settlement
- Returns logistics and warranty
- Tax engine and invoice generation internals
- Loyalty programme and ad-tech
- Production implementation of the full platform (design-first)

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User Stories & Acceptance Criteria

Given / When / Then for the critical flash-sale scenarios

ID	Story and acceptance criteria	Maps to	Verify
US-01	Last-item race. Given 1 unit remains, when two customers press Buy Now simultaneously, then exactly one receives a reservation and the other receives a deterministic SOLD_OUT response; stock never goes below 0.	INV-01, G-01	S, T
US-02	10,000 vs 100. Given 100 units and 10,000 concurrent attempts, when the sale opens, then at most 100 reservations exist at any instant, all others get SOLD_OUT or WAITING_ROOM, and no request is lost without a defined outcome.	NFR-01, G-01	S, L
US-03	Payment success. Given a valid reservation, when payment is captured, then the reservation becomes CONFIRMED, an order is created exactly once and the customer is notified.	FR-09, FR-11	T, D
US-04	Payment failure. Given a reservation in PAYMENT_PENDING, when the gateway declines, then the reservation is RELEASED, stock returns to available and the customer sees a clear failure state.	FR-13	T, D
US-05	Duplicate click. Given a Buy request with idempotency key K, when the same K is received again (double click, retry, replay), then the original result is returned and no second reservation, payment or order is created.	FR-07, FR-10	T, S
US-06	Payment timeout. Given a payment call times out, when the system retries, then it first queries gateway status by merchant reference and never charges twice.	FR-14, PAY-04	F, T
US-07	Paid but order service down. Given payment captured and Order Service unavailable for 30 s, when it recovers, then the order is created within 60 s, with no duplicate and no lost payment.	FR-15, NFR-21	F, S
US-08	Late payment. Given a reservation expired while 3-D Secure was in progress, when payment later succeeds, then stock is re-acquired if available; otherwise payment is auto-refunded and the customer is told.	RSV-07	T, S
US-09	Fair access. Given a customer has reached the per-sale limit, when they try again, then the request is rejected with LIMIT_REACHED without touching inventory.	INV-05	T
US-10	Ops control. Given anomalous behaviour, when the operator triggers the kill switch, then Buy Now is disabled within 10 s and in-flight reservations drain safely.	OPS-02	D, F

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Functional Requirements

v1.0 requirements retained and extended for sale operations, fairness and refunds

ID	Requirement	Priority	Verify
FR-01	Product Discovery. Customers discover products and sale/deal information; sale pages show live stock state (IN_STOCK / LOW / SOLD_OUT).	MUST	T
FR-02	Cart Management. Add, update and remove cart items. Adding to cart does <u>not</u> reserve stock.	MUST	T
FR-03	Inventory Check. Determine whether sufficient inventory exists before reservation (advisory read; the store of record decides).	MUST	T
FR-04	Inventory Reservation. Atomically and temporarily reserve stock for a valid Buy request, returning reservation_id and expires_at.	MUST	S
FR-05	Reservation Expiry. Release inventory automatically when a reservation expires.	MUST	S, F
FR-06	Reservation Release. Release on payment failure, cancellation or invalidation.	MUST	T
FR-07	Duplicate Reservation Prevention. Repeated Buy requests with the same idempotency key never create a second reservation.	MUST	T, S
FR-08	Checkout. Initiate checkout only for a valid, unexpired reservation owned by the caller.	MUST	T
FR-09	Payment Processing. Process payment and persist payment state transitions.	MUST	T
FR-10	Payment Idempotency. Repeated payment requests resolve to one transaction.	MUST	T, S
FR-11	Order Creation. Every captured payment eventually yields exactly one valid order.	MUST	F, S
FR-12	Order Lifecycle. Orders move only through valid transitions (Section 5.2).	MUST	T
FR-13	Payment Failure. Failure releases reservation and updates customer-visible state.	MUST	T
FR-14	Payment Timeout. Timeout triggers status query / reconciliation, never a blind re-charge.	MUST	F
FR-15	Order Recovery. Payment success + Order Service failure is recovered by event/retry/reconciliation.	MUST	F
FR-16	Shipment & Fulfilment. Confirmed orders enter fulfilment and shipment workflows.	SHOULD	T
FR-17	Notification. Generate customer events for reservation, payment, order and shipment milestones.	SHOULD	T
FR-18	Delivery Tracking. Support tracking after shipment via partner events.	SHOULD	T
FR-19	Sale Configuration (new). Admins define sale window, stock allocation, price, per-customer limit, reservation TTL and eligibility via RBAC-protected, audited APIs.	MUST	T, R
FR-20	Pricing & Promotions (new). Price, deal and coupon rules are evaluated server-side and locked at reservation time; client-supplied prices are never trusted.	MUST	T
FR-21	Cancellation (new). Customers may cancel until PROCESSING; stock returns to available and refund is initiated automatically.	SHOULD	T
FR-22	Refund (new). System supports full refund and void for late payment, cancellation and order-creation failure.	MUST	T, F
FR-23	Order Status Visibility (new). Customers see current order and payment status at all times, including an explicit “payment received, order being confirmed” state.	SHOULD	D
FR-24	Sale Controls (new). Operators can pause a sale, disable Buy Now, force sold-out and adjust stock, all audited.	MUST	D

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Lifecycle Models

Reservation and order are separate state machines, linked by payment

5.1 Reservation lifecycle (inventory side)

HAPPY PATH (BRIEF)



FAILURE PATH A (BRIEF)



FAILURE PATH B (BRIEF)



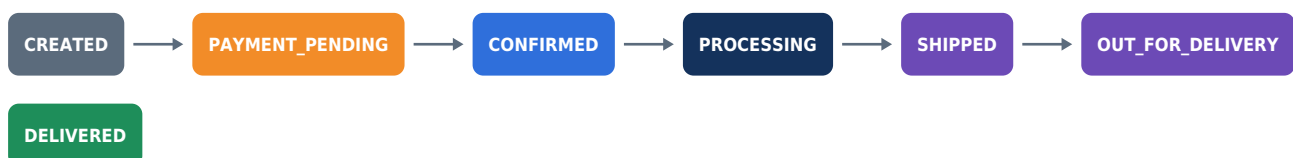
EDGE PATH C (NEW): PAYMENT AFTER EXPIRY



ID	Reservation rule	Priority	Verify
RSV-01	Reservation states and legal transitions are defined in an explicit table; illegal transitions are rejected and logged.	MUST	T
RSV-02	Each reservation carries reservation_id, customer_id, product_id, qty, idempotency_key, status, expires_at, version, created_at.	MUST	R
RSV-03	TTL : configurable per sale; default 10 min (TA); starts at reservation creation; measured on a single trusted server clock source.	MUST	T
RSV-04	TTL extension : one extension (default +5 min, TA) when a payment attempt is in progress (3-D Secure / OTP).	SHOULD	T
RSV-05	Expiry and release are idempotent and safe under concurrent executors; a reservation can be released at most once.	MUST	S
RSV-06	Expiry sweeper lag ≤ 30 s past expires_at (TA); a reservation may not block stock beyond TTL + 30 s.	MUST	S, M
RSV-07	Late payment rule : payment captured after expiry → attempt to re-acquire stock; if unavailable → automatic refund and notification. Outcome is deterministic and audited.	MUST	T, S
RSV-08	Reservation and payment state are cross-linked by reservation_id and merchant transaction reference.	MUST	R

5.2 Order lifecycle (commerce side)

HAPPY PATH (BRIEF)



FAILURE PATH



CANCELLATION PATH



RECOVERY PATH (PAYMENT SUCCESS, ORDER FAILURE)



ID	Order rule	Priority	Verify
ORD-01	An explicit transition table governs orders. Terminal states: DELIVERED, CANCELLED, REFUNDED, FAILED_RECONCILIATION (manual queue). Illegal transitions are rejected.	MUST	T
ORD-02	Order creation is exactly-once in effect via a unique constraint on payment reference / reservation_id.	MUST	T, S
ORD-03	Every state change that must publish an event uses a transactional outbox (or equivalent) so no event is lost or phantom.	MUST	R, F
ORD-04	Cancellation allowed until PROCESSING; afterwards only via return flow (out of scope). Cancellation returns stock and triggers refund.	SHOULD	T
ORD-05	Delivery partner integration sits behind an abstraction; adding a partner does not change order core.	SHOULD	R
ORD-06	Order history is append-only: every transition stores actor, reason, timestamp, before/after state.	MUST	T

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Inventory & Concurrency Requirements

The most consistency-critical part of SALESTORM

6.1 Invariants (strict guarantees)

ID	Invariant / rule	Priority	Verify
INV-01	Successful sales never exceed available inventory (Brief). 100 units → ≤ 100 sold, always.	MUST	S, L
INV-02	Conservation: available + reserved + sold = total_stock at every committed state; inventory is never negative.	MUST	S, M
INV-03	The durable store of record is the sole authority. Caches, counters and queues are advisory and never the final arbiter.	MUST	R
INV-04	Duplicate purchase requests never create duplicate reservations (idempotency key + uniqueness constraint).	MUST	T, S
INV-05	Per-customer purchase limit (default 1 unit per sale per account, TA), enforced server-side atomically with reservation.	MUST	T
INV-06	Expired reservations stop blocking stock; released stock becomes available to subsequent valid purchases.	MUST	S
INV-07	Cheap rejection: once sold-out, the ~9,900 losing requests are rejected from an edge/cache signal without touching the inventory store.	MUST	L
INV-08	Inventory adjustments (restock, cancellation return, expiry release) are recorded in an append-only ledger with actor and reason.	MUST	T, R
INV-09	Soft vs hard sold-out: SOFT_SOLD_OUT (stock held by pending reservations, may return) is distinguished from HARD_SOLD_OUT (sold = total). Cache TTL for soft state is short (≤ 2 s, TA).	SHOULD	S
INV-10	Transaction boundaries for reserve, confirm, release and expire are explicitly defined; each is a single atomic unit on the store of record.	MUST	R
INV-11	A comparison of ≥ 2 concurrency-control approaches (e.g. optimistic versioning vs pessimistic locking/atomic decrement) with justified selection is delivered in the ADR.	MUST	R

6.2 Concurrency-critical data (from Brief)

Field	Purpose	Field	Purpose
inventory_id	Unique inventory record	version	Concurrency / version control
product_id	Product reference	updated_at	Last update timestamp
available_quantity	Units available for reservation	reservation_id	Unique reservation reference
reserved_quantity	Units currently reserved	expires_at	Reservation expiry time
sold_quantity	Units confirmed as sold	idempotency_key	Duplicate request prevention

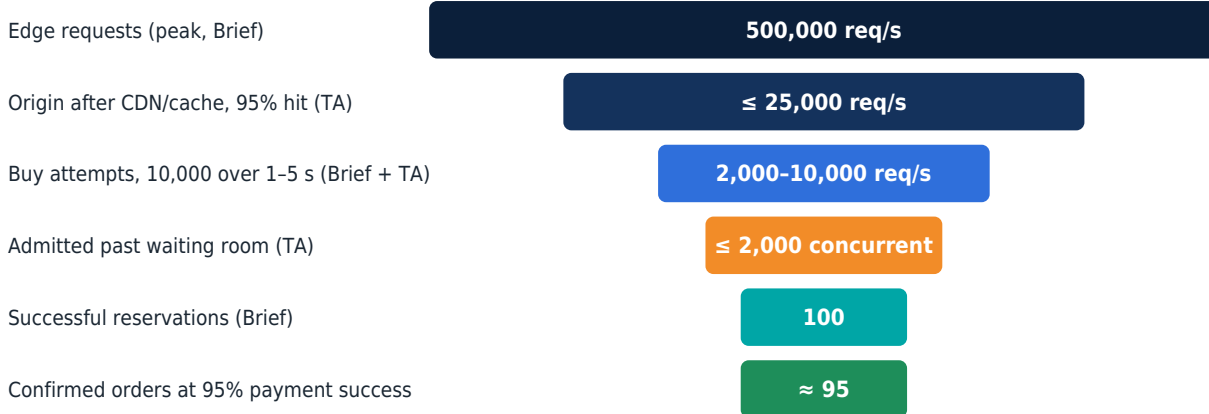
The database design must address primary keys, foreign keys, indexes, constraints (including CHECK available ≥ 0 and UNIQUE(idempotency_key)), transaction boundaries, concurrency control, consistency and audit information.

7

Traffic Model & Capacity Worksheet

Derived numbers that expose where the real bottleneck sits

Not all traffic is equal: ~95% of the 500,000 req/s is cacheable browse traffic. The contended, correctness-critical path is a tiny fraction — and within it, **one hot inventory record** is the true serialization point. All figures below marked TA are team assumptions.



ID	Worksheet item	Derived value	Status
CAP-01	Normal traffic → flash multiple	10,000 → 500,000 req/s = exactly 50× (answers the 50× question)	Brief
CAP-02	Cacheable share of peak traffic	≥ 95% (catalogue, sale page, static, stock-state signal)	TA
CAP-03	Dynamic origin load at peak	≤ 25,000 req/s	Derived
CAP-04	Buy-Now attempts and duplicates	10,000 attempts; 2% duplicates = 200 replays that must be idempotent	Brief
CAP-05	Reservation outcomes	100 winners; 9,900 SOLD_OUT / WAITING_ROOM; 0 undefined	Derived
CAP-06	Payment outcomes	95 captured, 5 failed → 5 units released and re-offered (soft sold-out reversal)	Brief
CAP-07	Edge bandwidth	~20 KB avg cached payload × 500k = ~10 GB/s (~80 Gbps), absorbed by CDN	TA
CAP-08	Log/telemetry volume	~1 KB per dynamic request × 25k/s = ~25 MB/s; 1-hour sale ≈ 90 GB (sampling for browse)	TA
CAP-09	Sale window	Burst: first 5 s; sustained: ~60 min; long tail: payment/orders for ≥ 24 h	TA
CAP-10	Hot-key contention	1 inventory record per product is the serial point; throughput of that point, not fleet size, bounds reservations	Derived

ID	Capacity requirement	Priority	Verify
CAP-11	The system distinguishes traffic overload (HTTP 429/503, retry-after) from business exhaustion (HTTP 409/410 SOLD_OUT) in every client response.	MUST	T
CAP-12	Stateful contention on the hot inventory record must not limit horizontal scaling of stateless tiers.	MUST	R
CAP-13	Design states where requests are throttled, queued and rejected , and the exact point where inventory contention is controlled.	MUST	R, D
CAP-14	At 50× growth (or 500k → 25M req/s) the document names which tiers scale out, which are re-sharded, and which assumptions break.	MUST	R

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Non-Functional Requirements (Quantified)

v1.0 NFR-01...14 retained; NFR-15...24 added with measurable targets

ID	Attribute	Requirement and measure	Type
NFR-01	Concurrency	Handle 10,000 simultaneous purchase requests for 100 units with controlled contention and a defined outcome per request.	Guarantee
NFR-02	Inventory consistency	Successful sales never exceed inventory. <i>Canonical: INV-01, INV-02, G-01.</i>	Guarantee
NFR-03	Scalability	Stateless tiers scale horizontally; capacity added without downtime.	Target
NFR-04	Normal traffic	Sustain ~10,000 req/s steady state.	Target
NFR-05	Flash traffic	Sustain 500,000 req/s at edge; origin $\leq 25,000$ req/s dynamic.	Target
NFR-06	Idempotency	Repeated requests and messages never create duplicate business transactions.	Guarantee
NFR-07	Reliability	Defined recovery path for each service, DB and gateway failure (Section 15).	Guarantee
NFR-08	Reservation correctness	Unpaid/expired reservations released. <i>Canonical: RSV-05, RSV-06.</i>	Guarantee
NFR-09	Payment safety	No duplicate payment transaction. <i>Canonical: PAY-02, G-04.</i>	Guarantee
NFR-10	Order correctness	Every successful purchase reaches a valid order state.	Guarantee
NFR-11	Fault tolerance	Timeouts, retries, circuit breakers, DLQ and workflow compensation designed per dependency.	Guarantee
NFR-12	Security	See Section 13 (SEC-01...SEC-10).	Guarantee
NFR-13	Observability	See Section 14 (OBS-01...OBS-08).	Target
NFR-14	Operability	Understandable and implementable by an engineering team; runbooks exist.	Target
NFR-15	Edge capacity (TA)	Edge sustains 500k req/s; $\geq 95\%$ served from CDN/cache.	Target
NFR-16	Buy latency (TA)	Reservation decision (success or sold-out) p99 ≤ 300 ms at API gateway, p50 ≤ 80 ms, excl. client network.	Target
NFR-17	Read latency (TA)	Product/sale page p99 ≤ 200 ms on cached path.	Target
NFR-18	Availability (TA)	Browse $\geq 99.95\%$; checkout/payment/order $\geq 99.9\%$ during sale window.	Target
NFR-19	Error rate (TA)	5xx $< 0.1\%$ under load. Business rejections (sold out, rate limited) excluded and tracked separately.	Target
NFR-20	Durability / DR (TA)	RPO = 0 for inventory, reservation, payment, order commits. RTO ≤ 5 min single-AZ; ≤ 30 min regional.	Guarantee
NFR-21	Order recovery (TA)	After Order Service recovery, 100% of paid-unordered payments converted ≤ 60 s; queue backlog drained ≤ 5 min.	Target
NFR-22	Elasticity (TA)	Peak capacity pre-warmed before sale start; autoscaling is a safety net, not the primary mechanism.	Target
NFR-23	Continuous correctness (TA)	Automated reconciliation of inventory, payments and orders every 5 min during the sale and daily after; zero unexplained diffs.	Guarantee
NFR-24	Reservation expiry (TA)	Released within TTL + 30 s; sweeper keeps lag below 30 s at p99.	Target

HONEST NOTE ON NUMBERS

The Brief specifies traffic but no latency SLA or availability percentage. Every value tagged (TA) above is a deliberate team assumption, tunable at HLD review; it does not alter the Brief's strict guarantees.

9

Traffic Protection, Fairness & Abuse Control

Flash sales are won or lost at the front door

ID	Requirement	Priority	Verify
TRF-01	Admission control / virtual waiting room caps concurrent entrants to the checkout path; admitted users hold a signed, short-lived token.	MUST	L
TRF-02	Rate limits per IP, per account and per device fingerprint; separate limits for browse and Buy Now.	MUST	T, L
TRF-03	Bot mitigation: WAF rules, challenge/CAPTCHA at Buy Now, anomaly detection on velocity and fingerprint reuse.	MUST	R, T
TRF-04	Load-shedding priority : protect checkout, payment and order above browse and recommendations.	MUST	L, F
TRF-05	Graceful degradation : reviews, recommendations and non-critical widgets can be disabled by feature flag during peak.	SHOULD	D
TRF-06	Responses distinguish 429/503 (overloaded, retry later) from 409/410 (sold out). Clients never retry on SOLD_OUT.	MUST	T
TRF-07	Queue position and approximate wait are shown to waiting-room users; fairness is first-admitted, not first-clicked, to avoid advantaging bots.	SHOULD	D
TRF-08	Request validation at the edge: authenticated session, valid sale token, bounded payload, schema validation before reaching services.	MUST	T
TRF-09	Retry guidance: clients use jittered exponential backoff; servers return Retry-After; retry storms are bounded.	SHOULD	L

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Payment Requirements

Industry-grade payment safety, reconciliation and compliance

10.1 Scenario matrix (from Brief, extended)

Scenario	Required outcome	Controls
Payment succeeds	Confirm payment, reservation → CONFIRMED → SOLD, create order	Webhook + sync confirm, outbox event
Payment fails	Release reservation, order → PAYMENT_FAILED / CANCELLED, notify	Idempotent release, ledger entry
Payment times out	Query gateway by merchant reference; retry or release only after status known	Timeout, backoff, reconcile job
Duplicate payment request	Same idempotency key returns same result; one transaction reference	Unique key + gateway reference
Success, Order Service fails	Payment stays CAPTURED; event/retry/reconcile creates order; refund if unrecoverable	Outbox, DLQ, recovery job
Success after expiry (new)	Re-acquire stock or auto-refund, deterministically	RSV-07, PAY-06
Webhook duplicate / out of order (new)	Deduplicate by event id; apply only forward state transitions	PAY-03

ID	Requirement	Priority	Verify
PAY-01	PCI scope minimisation: card data never touches SALESTORM servers; hosted page or tokenisation (SAQ-A style). Only tokens and last-4 are stored.	MUST	R
PAY-02	Every payment attempt has a unique idempotency key and merchant transaction reference, also sent to the gateway.	MUST	T, S
PAY-03	Webhooks are signature-verified, deduplicated by event id, processed idempotently and tolerant of reordering.	MUST	T
PAY-04	On timeout, query gateway status or reconcile before any retry or release. Never blindly re-charge.	MUST	F
PAY-05	Support 3-D Secure / OTP flows whose latency can exceed normal TTL; reservation TTL extends once on payment-in-progress (RSV-04).	MUST	T
PAY-06	Refund and void capability for late payment, cancellation and order-creation failure; refunds are idempotent.	MUST	T
PAY-07	Daily reconciliation between gateway settlement report and internal payment ledger; unmatched items flagged to Finance.	MUST	R, S
PAY-08	Timeout, retry-with-backoff and circuit breaker policy defined per external dependency (gateway p99 timeout, max retries, open/half-open thresholds).	MUST	F
PAY-09	Payment state machine: INITIATED → AUTHORISED → CAPTURED / FAILED / TIMED_OUT / REFUND_PENDING → REFUNDED; transitions are forward-only and audited.	MUST	T
PAY-10	Payment router supports multiple gateways/providers behind one abstraction; failover is manual-approved or policy-driven (documented trade-off).	SHOULD	R
PAY-11	Amount, currency and order total are re-verified server-side against the locked price before capture.	MUST	T

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Fulfilment & Notification Requirements

From confirmed order to delivered parcel

ID	Requirement	Priority	Verify
ORD-07	Confirmed orders generate a fulfilment request asynchronously; fulfilment failure never rolls back a paid order — it retries or escalates.	MUST	F
ORD-08	Shipment events from partners are idempotent, ordered per shipment and map to order states (SHIPPED, OUT_FOR_DELIVERY, DELIVERED).	SHOULD	T
ORD-09	Delivery partner failure triggers retry, then fallback partner or manual queue; no order is silently stuck (stuck-order alert).	SHOULD	F
ORD-10	Notifications are event-driven and idempotent (one message per milestone); channel failure falls back or retries without blocking the order.	SHOULD	T
ORD-11	Notification content never includes full card or sensitive PII; templates are versioned.	MUST	R

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Messaging & Event Reliability

Asynchronous workflows without lost or duplicated business effects

ID	Requirement	Priority	Verify
MSG-01	Delivery semantics stated per channel: at-least-once delivery with idempotent consumers (effectively-once business effect).	MUST	R, S
MSG-02	Dead-letter queue for failed messages with alerting, replay tool and documented retention (≥ 7 days, TA).	MUST	F
MSG-03	Event schemas are versioned and backward compatible; consumers tolerate unknown fields.	SHOULD	R
MSG-04	Per-key ordering where required (all events for one order / reservation / payment share a partition key).	MUST	R
MSG-05	Each event has an owner service; consumer responsibilities are documented in the event catalogue.	MUST	R
MSG-06	Retries use exponential backoff with jitter and a bounded attempt count before DLQ.	SHOULD	F
MSG-07	Every event carries event_id, correlation_id, causation_id, aggregate_id, version and timestamp.	MUST	T

13

Security & Compliance Requirements

Defence in depth for money, identity and stock

ID	Requirement	Priority	Verify
SEC-01	AuthN/AuthZ : OAuth2/OIDC + JWT for customers; mTLS or signed service tokens between services; RBAC (least privilege) for admin operations.	MUST	T, R
SEC-02	Encryption : TLS 1.2+ in transit everywhere; AES-256-class encryption at rest for PII and payment references.	MUST	R
SEC-03	Audit log : append-only, tamper-evident; records who, what, when, and before/after state for inventory, payment, order and admin actions.	MUST	T, R
SEC-04	Secrets stored in a managed vault, rotated, never in code or config repositories.	MUST	R
SEC-05	PII : minimisation, masking in logs, retention and deletion policy aligned with applicable law (e.g. India DPDP Act 2023); consent recorded.	MUST	R
SEC-06	Replay and CSRF protection on Buy Now and payment callbacks (nonce/timestamp, signed tokens, SameSite).	MUST	T
SEC-07	Input validation and output encoding on every API; schema validation; protection against injection and mass assignment.	MUST	T
SEC-08	Rate limiting and abuse protection are enforced at edge <u>and</u> service level (defence in depth).	MUST	T, L
SEC-09	Dependency and image vulnerability scanning in CI; signed artefacts; threat model (STRIDE) maintained for the checkout path.	SHOULD	R
SEC-10	Admin and kill-switch actions require MFA and are dual-controlled for stock changes above a threshold.	SHOULD	R

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Observability & Operations

Detect, decide and act within seconds during the sale

ID	Requirement	Priority	Verify
OBS-01	SLIs and SLOs defined for NFR-16...NFR-19 with an error-budget policy.	MUST	R
OBS-02	A single correlation/trace ID spans request → reservation → payment → order → notification.	MUST	T
OBS-03	Required metrics: request rate, latency, error rate, reservation failures, payment failures, order conversion, queue backlog/lag.	MUST	M
OBS-04	Alerts (examples, TA): any available < 0 or conservation breach → page immediately; payment failure rate > 10% for 2 min; queue lag > 60 s; DLQ depth > 0; stuck order > 5 min.	MUST	F, M
OBS-05	Structured JSON logs for business events (reservation_created, reservation_expired, payment_captured, order_created, refund_issued) with correlation ID.	MUST	T
OBS-06	Business dashboard: stock remaining, active/expiring reservations, payment success rate, order conversion, DLQ depth, waiting-room length.	SHOULD	D
OBS-07	Distributed tracing sampled at 100% for checkout/payment/order, lower for browse.	SHOULD	R
OBS-08	Synthetic probes run the full purchase flow every minute during the sale.	COULD	M
OPS-01	Pre-sale testing : load test at ≥ 1.5× expected peak; failure injection for Order Service outage (30 s), gateway timeouts and database failover.	MUST	L, F
OPS-02	Runbooks and kill switch : pause sale, disable Buy Now, force sold-out, drain reservations; each tested in rehearsal.	MUST	D
OPS-03	Zero-downtime deployment with change freeze during the sale window; instant rollback path.	MUST	R
OPS-04	Feature flags control degradation levels (L0 normal → L3 checkout-only) with documented triggers.	SHOULD	D
OPS-05	Post-sale review: reconciliation report, SLO report, incident log and action items within 48 h.	SHOULD	R

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Availability, Failure & Disaster Recovery

Every failure has a named recovery path

ID	Failure	Required behaviour	Target
DR-01	Database primary failure	Automatic failover to synchronous replica; no committed inventory/payment/order lost; in-flight requests retry idempotently	RPO 0, RTO ≤ 5 min
DR-02	Availability-zone loss	Stateless tiers and data replicas span ≥ 3 AZs; traffic shifts automatically	RTO ≤ 5 min
DR-03	Regional outage	Cross-region restore/standby; sale paused or failed over under operator control	RTO ≤ 30 min
DR-04	Payment gateway outage	Circuit breaker opens; reservations extended once; checkout shows pending/maintenance; secondary provider if configured; reconcile on recovery	No double charge
DR-05	Order Service outage (30 s, Brief)	Payments stay CAPTURED; events queue durably; orders created on recovery	≤ 60 s after recovery
DR-06	Cache / counter loss	Falls back to store of record; no correctness impact, only latency	No data loss
DR-07	Message broker failure	Outbox retains events; relay resumes; consumers deduplicate	No lost events
DR-08	Notification / delivery partner outage	Retry, fallback channel/partner, never block orders	Eventual delivery
DR-09	Backup and restore	Point-in-time recovery; quarterly restore drill; backups encrypted	Drill passed

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Data Requirements & Governance

Entities, integrity rules and retention

Entity (Brief)	Integrity and governance requirement
CUSTOMER	PII classified; encrypted at rest; consent and deletion workflow (SEC-05)
PRODUCT / CATEGORY	Cacheable; versioned; price locked into reservation snapshot
INVENTORY	CHECK available ≥ 0 ; version column; conservation invariant verified by reconciliation
INVENTORY_RESERVATION	UNIQUE(idempotency_key); expires_at indexed for sweeper; status transitions audited
CART / CART_ITEM	Non-authoritative for stock or price; may be ephemeral
ORDER / ORDER_ITEM	UNIQUE(payment_ref / reservation_id); append-only history; retention $\geq$ legal minimum (TA: 8 years for financial records)
PAYMENT	UNIQUE(merchant_txn_ref); no PAN stored; ledger entries immutable
SALE / DEAL / COUPON	Effective-dated; changes audited; server-side evaluation only
SHIPMENT / NOTIFICATION	Idempotent event application; delivery state history retained
AUDIT / LEDGER (new)	Append-only, tamper-evident, time-ordered; separate retention and access policy
OUTBOX / DLQ (new)	Written in the same transaction as the state change; purge after confirmed publish

Database design must address PKs, FKs, indexes, constraints, transaction boundaries, concurrency control, consistency and audit (Brief).

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Assumptions, Constraints & Dependencies

Explicit so they can be challenged at review

17.1 Assumptions

ID	Assumption	Basis / Impact if wrong
A-01	Flash sale involves limited-stock products.	Brief. Core scenario.
A-02	Critical test: 100 units vs 10,000 concurrent requests.	Brief.
A-03	Payment depends on an external gateway.	Brief. Drives PAY-04, DR-04.
A-04	Downstream services can be slow or fail.	Brief. Drives NFR-11.
A-05	Both synchronous and asynchronous communication are used.	Brief. HLD must justify each.
A-06	Caching, queues, databases, monitoring, logging, tracing are available.	Brief. Reference architecture.
A-07	Numeric SLAs are team assumptions (marked TA).	Brief gives none. Revisit at HLD.
A-08	Prototype is validation evidence, not the deliverable.	Brief. Design-first.
A-09	Sale concerns one or a few hot SKUs; all 10,000 users target the same product.	Single hot record is the contention point (CAP-10).
A-10	Customers are authenticated before Buy Now.	Enables per-account limits and fairness.
A-11	Gateway offers idempotency keys, status query, webhooks and settlement reports.	If not, compensating polling and dedupe must be stronger.
A-12	Sale start time is known in advance, enabling pre-warming.	Enables NFR-22; unannounced spikes need aggressive autoscale.
A-13	Cloud-style infrastructure with ≥ 3 AZs and managed services is available.	Affects DR targets and operations burden.
A-14	Clock skew is bounded and expiry uses a single trusted time source.	Risk R-05.

17.2 Constraints

ID	Constraint
C-01	Must handle 10,000 concurrent requests / 100 units.
C-02	Must reason about up to 500,000 requests/sec.
C-03	Overselling is unacceptable.
C-04	Duplicate reservations, payments and orders must be prevented.
C-05	Reservations must support expiry and release.
C-06	Payment and order workflows need explicit failure and recovery paths.
C-07	Design-first; production implementation not required.
C-08	AI may accelerate prototypes, tests, simulations and docs but not requirements, architecture, concurrency, consistency, trade-offs or final explanation.
C-09	Every team member must understand the architecture and any AI-generated artefact.
C-10	(new) Card data must never be stored or processed on SALESTORM servers (PCI scope).
C-11	(new) Compliance with applicable data-protection law (e.g. DPDP Act 2023).

17.3 Critical dependencies

Dependency	Requirement impact	Failure plan
Payment Gateway	Success/failure/timeout, idempotency, webhooks, settlement	DR-04, PAY-04, PAY-08
Database	Inventory, reservation, order, payment durability and consistency	DR-01, NFR-20
Message Broker	Order recovery, async workflows, events	DR-07, MSG-01...07
Cache / CDN / WAF	Edge absorption, sold-out signal, bot control	DR-06, TRF-03
Shipment / Delivery Partner	Fulfilment and tracking lifecycle	ORD-09, DR-08
Notification Provider	Customer communication	ORD-10, DR-08
Identity Provider	Authentication and tokens	Token validity window, SEC-01

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Strict Guarantees vs. Performance Targets

What may never be violated vs. what we aim for

18.1 Strict guarantees (zero tolerance)

ID	Guarantee	Meaning	Detection
G-01	No overselling	100 units never yield more than 100 successful sales	Conservation check, alert page
G-02	No negative inventory	available_quantity ≥ 0 under any concurrency	DB CHECK constraint
G-03	No duplicate reservation	Same idempotency key never reserves twice	UNIQUE constraint
G-04	No duplicate payment	Repeated payment requests resolve to one transaction	Gateway ref + recon
G-05	Reservation release	Unpaid/expired reservations released exactly once	Sweeper + recon
G-06	Valid order state	Every captured payment reaches a terminal order state	Paid-without-order job
G-07	Defined recovery	Every failure has a documented, tested path	Failure-injection drills
G-08	No lost event	State changes and events commit atomically (outbox)	Outbox lag + DLQ

18.2 Targets (measured, error-budgeted)

Target	Value	Origin
Normal traffic	~10,000 req/s	Brief
Flash-sale edge traffic	500,000 req/s; ≥ 95% cached	Brief + TA
Buy-Now p99 latency	≤ 300 ms	TA
Browse availability	≥ 99.95%	TA
Checkout availability	≥ 99.9%	TA
Order recovery after outage	≤ 60 s, backlog ≤ 5 min	TA
50x traffic scenario	Documented architectural response	Brief

19**Risk Register**

Known threats to the requirements, resolved in HLD and ADR

ID	Risk	Likelihood / Impact	Mitigation direction (resolved in HLD)
R-01	Hot-row contention on the single inventory record	High / High	Atomic conditional update, admission control, edge sold-out signal, optional stock sharding with strict sum invariant
R-02	Payment gateway slowness or outage	Med / High	Timeouts, circuit breaker, TTL extension, reconcile, optional secondary provider
R-03	Cache shows stale stock or wrong sold-out state	High / Med	Cache is advisory only (INV-03); store of record decides; short soft-sold-out TTL
R-04	Bot / scalper traffic dominates stock	High / High	WAF, challenge, per-account limits, waiting room, device fingerprinting
R-05	Clock skew breaks reservation expiry	Low / Med	Single trusted time source in DB; expiry evaluated server-side
R-06	Retry storms amplify an outage	Med / High	Jittered backoff, Retry-After, load shedding, bounded retries
R-07	Lost or duplicated events	Med / High	Outbox, idempotent consumers, DLQ, replay
R-08	Paid but no order (silent)	Med / High	Recovery job, paid-without-order alert, FAILED_RECONCILIATION queue
R-09	Admin or insider error (wrong stock, wrong price)	Low / High	RBAC, MFA, dual control, audited ledger, preview + rollback
R-10	Thundering herd at sale start	High / Med	Pre-warm, staged admission, CDN pre-population
R-11	Late payment after expiry	Med / Med	RSV-07 deterministic re-acquire or refund

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Requirement-to-Verification Matrix

Every MUST maps to a test, simulation, review or drill

Scenario / Experiment	Requirements proven	Method	Pass criterion
Simulation: 10,000 threads vs 100 units, 2% duplicates, 5% payment failure	INV-01...04, FR-04, FR-07, CAP-04...06	S (AI-assisted harness)	sold $\leq$ 100; conservation holds; 0 duplicate reservations; 5 released units re-offered
Last-item race: 2 concurrent requests for 1 unit	US-01, G-01, G-02	S, T	exactly 1 winner, 1 SOLD_OUT; available never < 0
Idempotent replay: same key xN	US-05, FR-07, FR-10, PAY-02	T	single reservation/payment/order; identical response
Order Service down 30 s after payment success	US-07, FR-15, DR-05, NFR-21	F, S	all paid orders created $\leq$ 60 s after recovery, no duplicates
Gateway timeout then status query	US-06, PAY-04, PAY-08	F	no double charge; correct final state
Reservation expiry + late payment	RSV-05...07, US-08	S, T	release once; late payment re-acquires or refunds
DB failover during sale	DR-01, NFR-20	F	RPO 0; RTO $\leq$ 5 min; invariants intact
Load test at 1.5x peak	NFR-15...19, TRF-04	L	SLOs met; load shedding protects checkout
Reconciliation run	NFR-23, PAY-07, INV-02	S, R	zero unexplained diffs
Kill-switch drill	OPS-02, US-10	D	Buy Now disabled $\leq$ 10 s; reservations drain safely

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Success Criteria

The requirements are satisfied when the architecture demonstrates:

Area	Expected outcome	Measured by
Concurrency	10,000 simultaneous requests handled with controlled contention; every request has a defined outcome	Simulation, load test
Inventory	100 units never produce more than 100 sales; conservation holds	G-01, INV-02, reconciliation
Reservation	Unpaid/expired reservations released; late payment handled deterministically	RSV-05...07
Payment	No duplicate transaction; timeout handled safely; refunds work	PAY-02, PAY-04, PAY-06
Order	Every successful purchase reaches a valid terminal order state	G-06, NFR-21
Reliability	Recovery path for service, DB and gateway failure with RPO/RTO	DR-01...09, OPS-01
Scalability	Horizontal scale; clear 50x response; named bottleneck and mitigation	CAP-14, NFR-15
Security	AuthN/Z, TLS, rate limits, validation, PCI scope, audit, secrets, privacy	SEC-01...10
Observability	Metrics, logs, traces, actionable alerts, dashboards	OBS-01...08
Defensibility	Every major decision justified with trade-offs in ADR	02_HLD, 10_ADR

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Requirements-to-HLD Handoff

The HLD must answer these questions without changing the requirements

#	Question for 02_HLD	Linked requirements
1	Where does high-volume traffic enter and how is it distributed?	NFR-15, CAP-02, CAP-03
2	Where are requests throttled, queued or rejected?	TRF-01, TRF-02, CAP-13
3	Where exactly is inventory contention controlled?	INV-03, CAP-10, INV-10
4	How is inventory consistency guaranteed (and which of ≥ 2 approaches was chosen)?	INV-01, INV-02, INV-11
5	Which components scale horizontally, and what is the hot-key strategy?	NFR-03, CAP-12
6	Which interactions are synchronous and which asynchronous, and why?	A-05, MSG-01
7	How are timeout, duplicate payment and payment-success/order-failure recovered?	PAY-02, PAY-04, FR-15
8	How is late payment after reservation expiry handled?	RSV-07, PAY-06
9	Where are cache, queue, database, monitoring, logging and tracing used?	A-06, OBS-02
10	How does the cache stay advisory while sold-out rejection stays cheap?	INV-03, INV-07, INV-09
11	What are the major bottlenecks and mitigations, including 50x growth?	CAP-14, R-01, R-10
12	How are DR targets (RPO 0, RTO 5/30 min) achieved?	NFR-20, DR-01...03
13	How are fairness, bots and waiting room designed?	TRF-01...07
14	What trade-offs does the selected architecture introduce?	ADR

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Traceability to Hackathon Brief & Next Artifacts

Requirements are the contract for every later diagram

Brief expectation	Covered in	Next artifact
Requirement analysis, assumptions, constraints	Sections 1–3, 17	02_HLD
Functional requirements	Sections 3–5	02_HLD, 05_API
Throughput, latency, availability, consistency, security, recovery	Sections 7, 8, 13–15	02_HLD, 08_Scalability_Reliability
Strict guarantees vs targets	Section 18	10_ADR
10,000 requests / 100 units; reservation and concurrency	Sections 6, 7	03_LLD, 04_Database
Payment and order reliability	Sections 5, 10, 11, 12	03_LLD, 05_API, 08_Scalability_Reliability
Security and observability	Sections 13, 14	09_Security_Observability
Stress scenarios, 50x, failure recovery	Sections 7, 15, 20	08_Scalability_Reliability, 12_Presentation
AI-assisted validation (optional)	Section 20	11_AI_Assisted_Validation

DESIGN-FIRST PRINCIPLE

Requirements are the contract. Later diagrams, schemas and any AI-assisted prototype must trace back to the IDs in this document, especially the inventory consistency (G-01, G-02), payment safety (G-04) and failure-recovery (G-07) guarantees. Architecture decisions — SQL vs NoSQL, optimistic vs pessimistic control, queue and cache technology, service decomposition — are intentionally deferred to 02_HLD and 10_ADR.